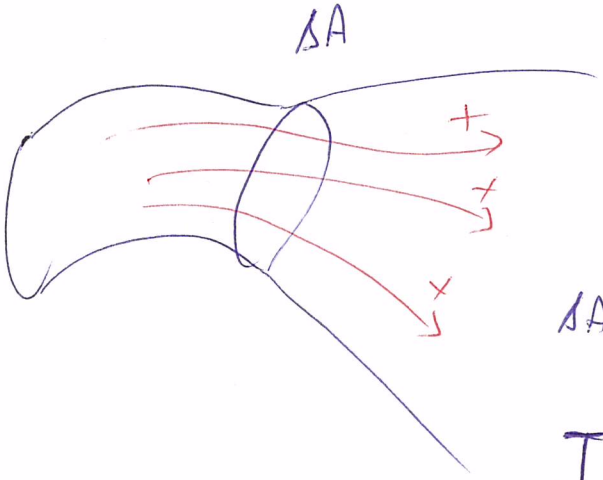
# 6 Electric current, conduction ⇕ convection

We have seen that in many materials charge can be travelling from one place to another.




The electric current is the quantity of charge going through the cross section $\Delta A$ of a conductor at a ~~given~~ unit time:

$$I_{avg} = \frac{\Delta Q}{\Delta t}$$

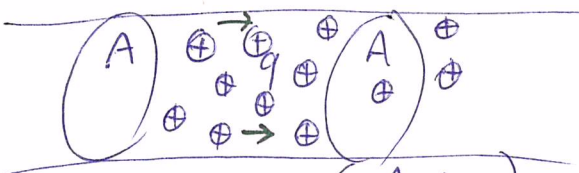
with $\Delta t \to \emptyset$ we can define instantaneous current

$$I := \frac{dQ}{dt} \qquad [I] = 1\,\frac{C}{s} = 1A$$

Convention: although $I$ is not a vector (it is rather a *flux* !!) it has a *flow* direction in the wire, conventionally the direction of movement of + charges

## Simple microscopic model:

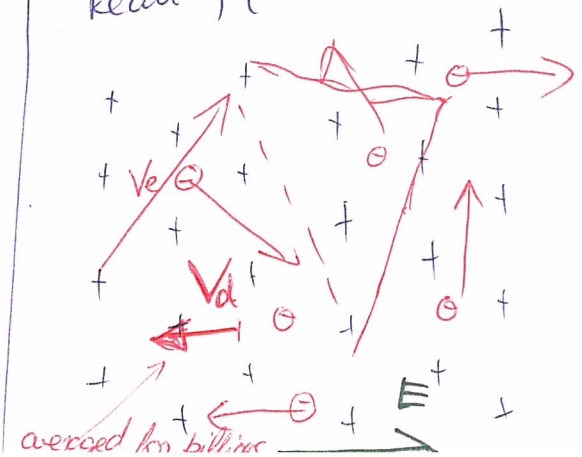
$$\Delta x = v_d \cdot \Delta t$$

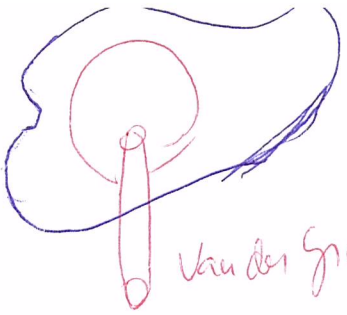



$$\Delta Q = (nA\,\Delta x)\,q$$

$$I_{avg} = \frac{\Delta Q}{\Delta t} = \frac{nA\,\Delta t \cdot v_d\, q}{\Delta t} = n q v_d A$$

Reality (metallic conductor):

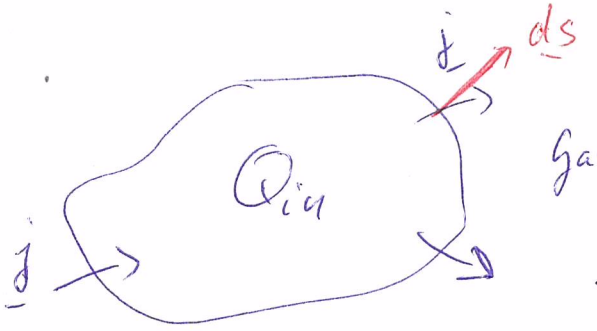





$$\oint \underline{j}\, d\underline{S} = ? - \frac{dQ_{in}}{dt}$$

conservation of charge!

Continuity equation!




Gauss-O $\oint \underline{j}\, d\underline{s} + \frac{dQ_{in}}{dt} = (\text{Source} \neq 0)$

$$\iiint \operatorname{div} \underline{j}\, dV + \iiint \varrho_{in}\, dV = 0$$

$$\boxed{\operatorname{div} \underline{j} = -\frac{\partial \varrho}{\partial t}}$$

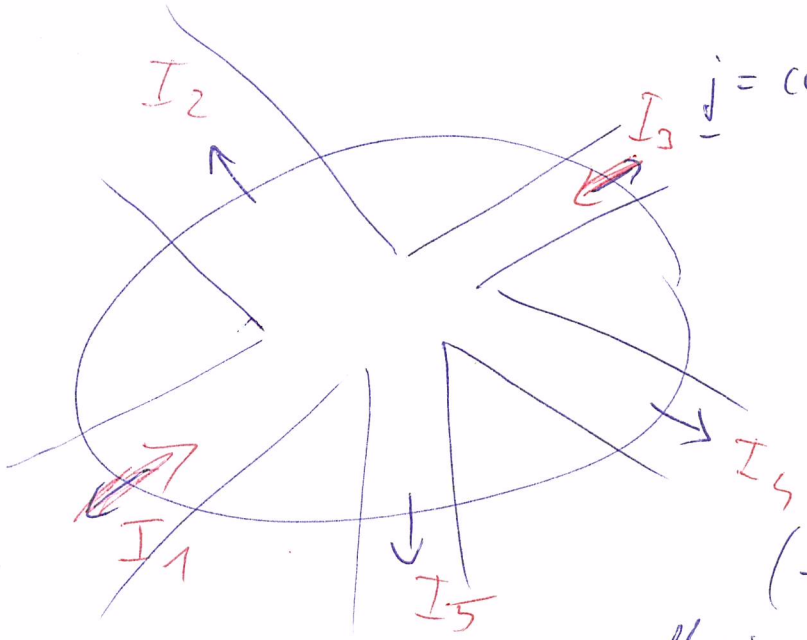
Let's elaborate our microscopic model a little more!

electron gas:

Let's apply this to stationary flow

(DC: direct current)

$\underline{j}$ = const in time, $\frac{\partial Q}{\partial t} = 0$




$$\oint \underline{j}\, d\underline{s} = 0$$

$$\| $$

$$I_1 + I_2 + I_3 + \ldots = 0$$

the sum of the currents

(+ flow out − flow in)

flowing in and out to a junction is 0

Kirchhoff's first law

Let us define the current density:

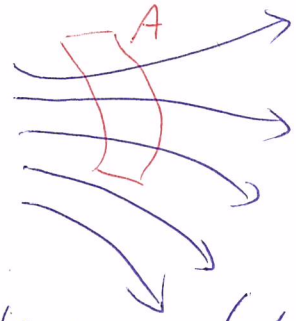
$$j := \frac{I}{A} = nqv_d$$

This can be made a vector!

$$\underline{j} := nq\underline{v}_d \qquad [j] = \frac{A}{m^2}$$

The surface integral (flux) of $\underline{j}$ is $I$

$$I = \iint_A \underline{j}\, d\underline{A}$$

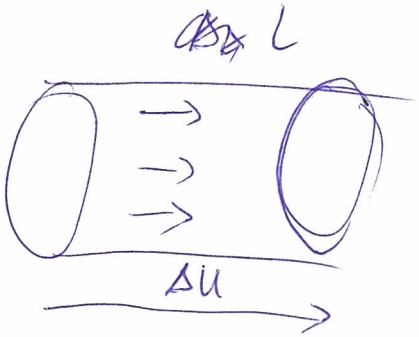



Inhomogenous current

Observation of Georg Simon Ohm: (1827)

$\underline{j} \sim \underline{E}$ simply $\underline{j} = \sigma \underline{E}$ Ohm's law

($\sigma$: conductivity of the material)




for a homogenous current:

$$I = j \cdot A = \sigma E A = \sigma A \frac{\Delta U}{l}$$

$$\Delta U = E \cdot l$$

pot. diff. or voltage

$$I = \frac{\sigma A}{l} \cdot \Delta U$$

$$\Delta U = \frac{l}{\sigma A} I = R I$$

R Resistance

$$[R] = 1\,\frac{V}{A} = 1\,\Omega \qquad [\sigma] = \frac{1}{\Omega m}$$

($\rho = \frac{1}{\sigma}$ resistivity, $\Omega m$, we will not use)

good insulator — good conductor

$\sigma$ varies in a wide range $10^{-18}\,\frac{1}{\Omega m} \rightarrow 10^{9}\,\frac{1}{\Omega m}$

Let's elaborate our microscopic model a little more!

① $E=0$

plenty ($10^{23}$) of free electrons having a thermal movement and colliding with atoms and each-other: electron gas ($v_{th} \cong 10^6 \frac{m}{s}$)

$N(10^{23})$

$$v_{ave} = \sum_i v_{th,i} / N = 0 \text{ !} \quad \text{no drift} \quad E = 0$$

② We switch on E

- $e^-$-s accelerate due to E between two collisions.
- next collision: extra energy of $e^-$ is transfered to the atom (thermalizing) → temperature increases!

$$v_{final} = v_{th} + \frac{qE}{m_e} \cdot t_\tau \qquad \tau: \text{average time between collisions}$$

$$v_{final\,ave} = \sum v_{th} + \frac{qE}{m_e} \cdot \tau = 0 + \frac{qE}{m_e}\tau = v_D$$

(Fizeau Bunsen Faraday Galileo Gödgös Bence Coulomb kis.)

$(v_D \sim 10^{-4} \frac{m}{s})$

$$v_D = \frac{q\tau}{m_e} E = \mu_e E$$

Drude model (Paul Drude 1900)

$\mu_e$: carrier mobility el. hole etc.

$$j = n q v_D = \frac{n q^2 \tau}{m_e} E \quad (= n q \mu_e E)$$

$$\sigma = \frac{n q^2 \tau}{m_e}$$

$\sigma = n q \mu_e$

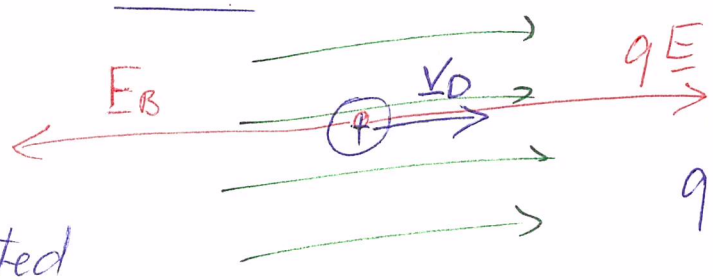
classical model, explains Ohm's law, not good for temp. description

# Energy & Power in el. currents

Scattering of electrons is a breaking or dissipative force

~~Its power:~~

Model: $\underline{E}$




$$q\underline{E} + \underline{F}_B = 0$$

The dissipated power $P_{1e^-} = \underline{F} \cdot \underline{v}$ $\left( \frac{dW}{dt} = \frac{\underline{F} \cdot d\underline{s}}{dt} = \underline{F} \cdot \underline{v} \right.$

$$P_N = N P_1 = nV \cdot q\underline{E} \cdot \underline{v} = \underbrace{nq\underline{v}}_{\underline{j}} \cdot \underline{E} \cdot V$$

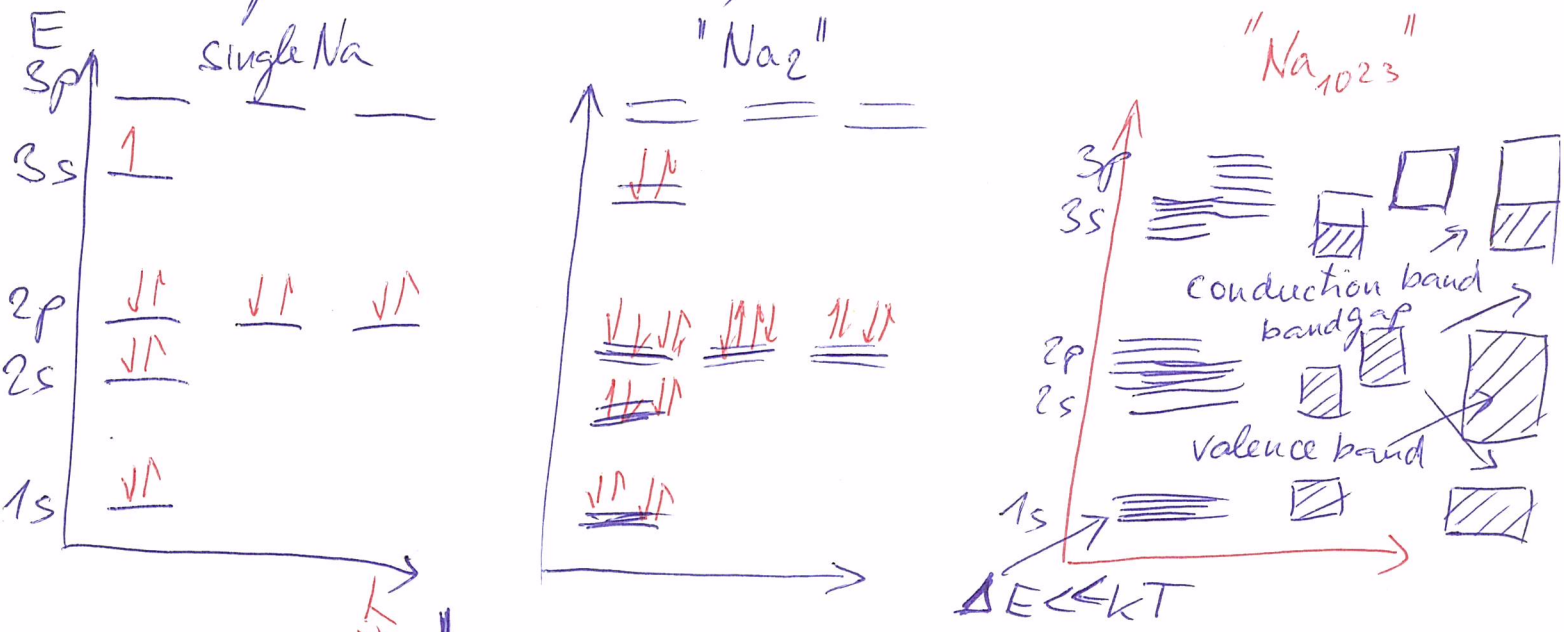
$$= \underline{j}\,\underline{E} \cdot V$$

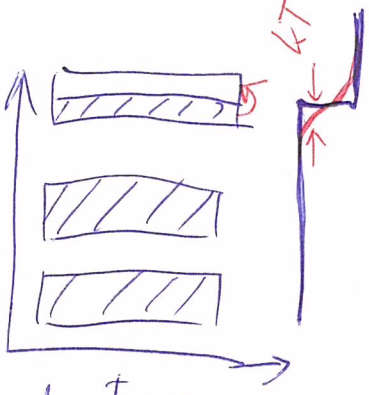
$$P = \underline{j}\,\underline{E}\,V = \frac{I}{A} \cdot \frac{U}{l} \cdot V = U \cdot I$$ ← Joule heat!

# Conduction in solids

## I Crystalline solids with metallic or covalent bond

Example: Na (Sodium) it has 11 $e^-$s



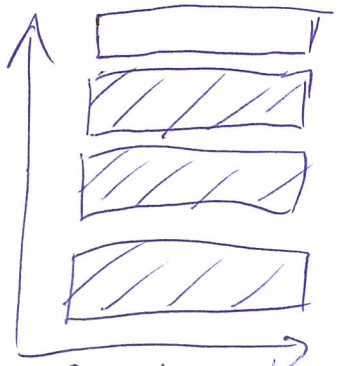



Conductors:
Half full/partially populated conduction band
free electrons

$$\sigma = \frac{n q^2 \tau}{m_e} = n q \mu_e$$

$T\uparrow \rightarrow \tau\downarrow$ (Moving atoms)
$\rightarrow \sigma\downarrow$

$\sigma = 10^7 - 10^9 (\Omega m)^{-1}$

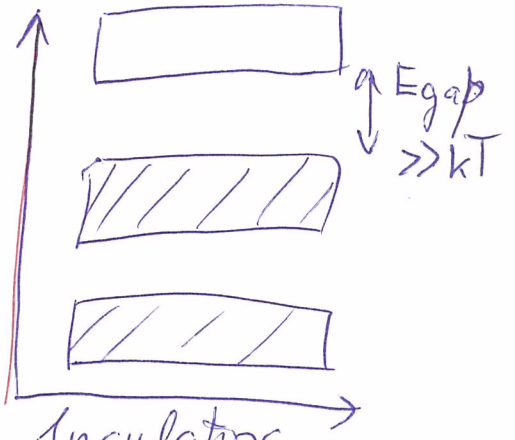


Semicond.
Empty cond.
$E_{gap} \sim kT$

$$\sigma = \frac{n q^2 \tau}{m_e}$$

$T\uparrow \sigma\uparrow$

$\sigma = 10^{-7} - 10^5$




Insulators
Empty cond. band,
$E_{gap} \gg kT$

$$\sigma = \frac{n q^2 \tau}{m_e}$$

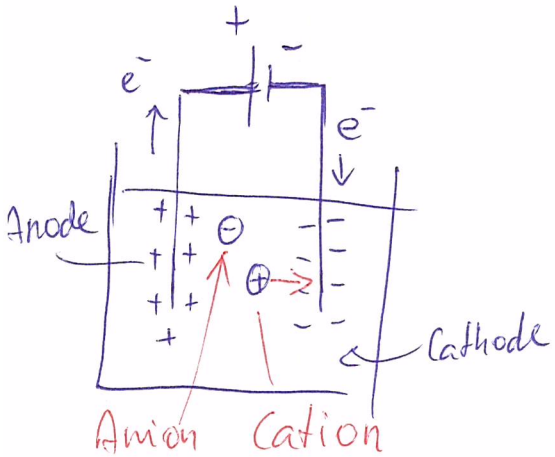
$\sigma = 10^{-8} - 10^{-18}$

(II) Ionic crystals : ionic conduction very low, good insulators

(III) Non-crystalline solids : no band structure, very low ionic or e⁻ conduction

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## Conduction in ~~solids~~ liquids :




Insulators, unless some ions are added. → called electrolites

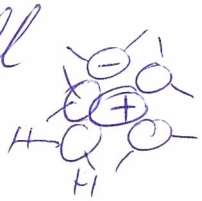
conduction processes : electrolitical pr.

\+ ions (tipically metal) deposit on − electrode → electro-galvanization

(Protection of ~~Fe~~ iron from corosion)

Conductivity lower than that of metals : lower mobility, higher mass, ions in ~~water~~ ~~bubble~~ shell

dipole



$$\sigma = \sum_i q_i n_i \mu_i$$ (non-linear with concentration!)

depends on viscosity of fluid, T ↑ visc ↓ σ ↑

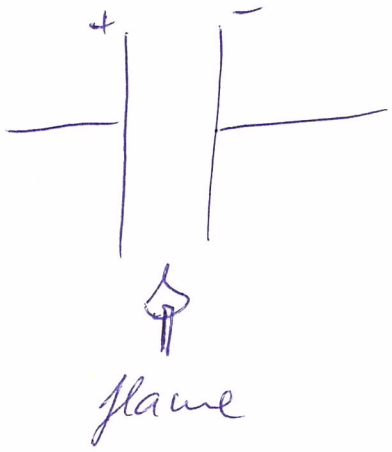
| | |
|---|---|
| Destilled water | $10^{-6}$ 1/Ωm |
| seawater | ~5 1/Ωm |
| 20% HCl | ~ 76 1/Ωm |

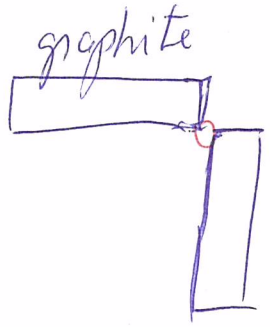
# Conduction in gases :

Gases are good insulators at atmospheric conditions and low field ($\sigma \sim 10^{-14} \frac{1}{\Omega m}$)

Ionization < externally induced (heat, radiation)
internally ionised (effect of ~~current~~ accelerated charge)

## 1. External ionization :

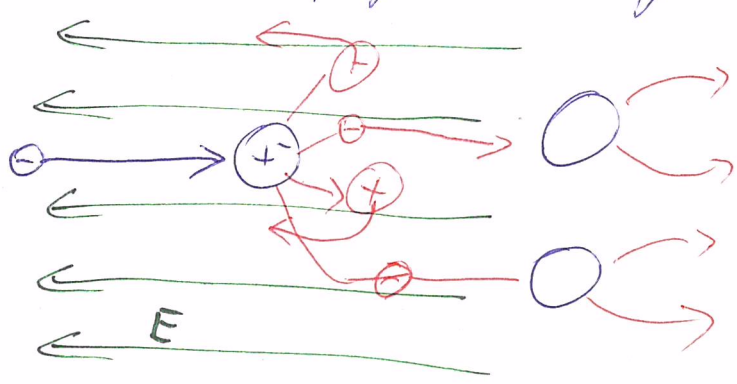



electric arc (heat)
$e^-$

## 2. Internal ionization

Some ion and $e^-$ is always present (cosmic radiation)
↳ can multiply in strong field

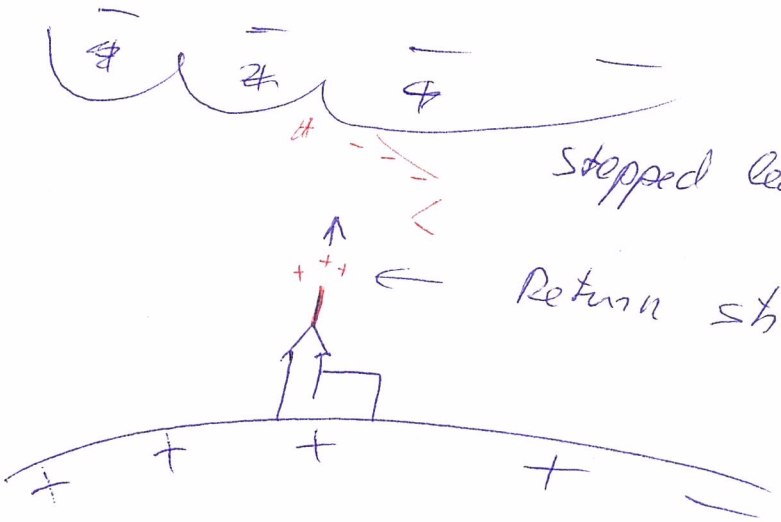



Ionization channel is formed.

$E_{threshold}$ :
$10 kV/cm$ $(1 \frac{MV}{m})$

Lower pressure : lower threshold in dry air !
why ?

# Lightning, electrical safety




lightning rod { Story (Prokop Divis ~~ch~~ Czech priest 1754
Benjamin Franklin ~ 1752-60.

Safest position in lightning:



There is an argument

Best: go to a covered place

Bird on the wire!

No text
① (house, car etc)
② Low land
③ Under a forest

↗ (Not under a single tree)

3-5 % direct shike

50% ground shike